University of Waterloo

Department of Electrical and Computer Engineering

ECE 457-A Adaptive and Cooperative Algorithms

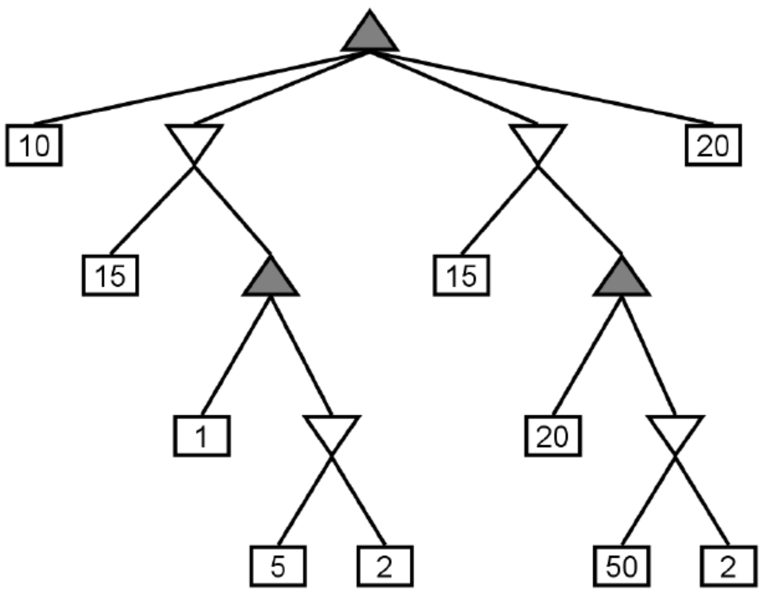
Midterm Examination Spring 2013

Time 17:30-18:50

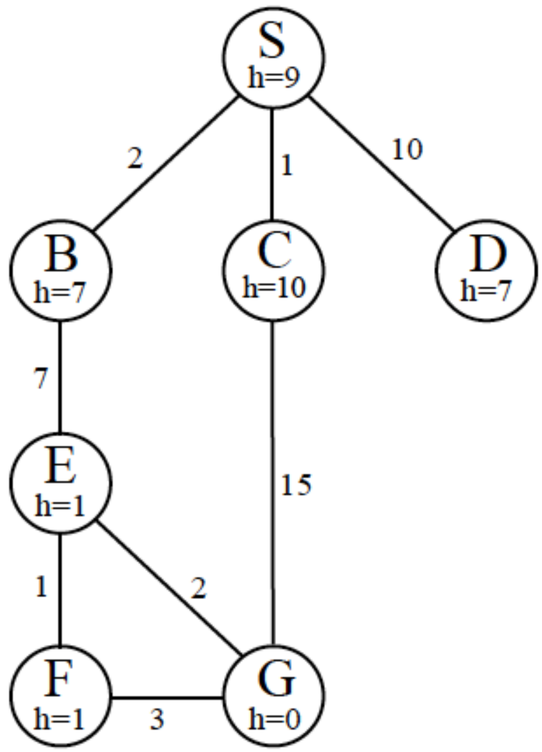
PLEASE ATTEMPT ALL QUESTIONS

Question 1. [25 Points] Search Techniques

a. For the following tree answer the following questions
   i. What is the minimax value for the root?
   ii. Draw an X through any nodes which will not be visited by alpha-beta pruning, assuming children are visited in left-to-right order.
   iii. Is there another ordering for the children of the root for which more pruning would result? If so, state the order.



b. Consider the search graph shown below. S is the start state and G is the goal state. All edges are bidirectional.
   For each of the following search strategies, give the path that would be returned, or write none if no path will be returned. If there are any ties, assume alphabetical tiebreaking (i.e., nodes for states earlier in the alphabet are expanded first in the case of ties).
   i. Depth-First graph search
   ii. Breadth-First graph search
   iii. Uniform cost graph search
   iv. Greedy graph search
   v. A* graph search



Question 2. [25 Points] Tabu Search

a. Define tabu tenure. What impact does it have on search performance? What is the rule of thumb for choosing a value for this control variable?

- Tabu tenure: the length of time *t* for which a move is forbidden

  t too small - risk of cycling

  t too large - may restrict the search too much

The rule of thumb:

$$t = \sqrt{n}$$

b. Explain the use of concepts such as Aspiration and Long Term Memory in Tabu Search.

- Aspiration criterion: Tabus may sometimes be too powerful: they may prohibit attractive moves, even when there is no danger of cycling, or they may lead to an overall stagnation of the searching process. It may, therefore, become necessary to revoke tabus at times. The criterion used for this to happen in the present problem of TSP is to allow a move, even if it is tabu, if it results in a solution with an objective value better than that of the current best-known solution.
- Tabu incorporates two type of memory: long term memory and short term memory. Short term memory stores more recent moves and long term memory keeps all the related moves.
  Short-term: The list of solutions recently considered. If a potential solution

appears on this list, it cannot be revisited until it reaches an expiration point. Long-term: Rules that promote diversity in the search process (i.e. regarding resets when the search becomes stuck in a plateau or a suboptimal dead-end).

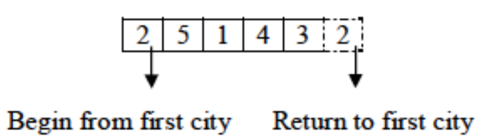
c. Consider the following TSP (Travelling Salesman Problem). There are 5 cities: C1,C2,C3,C4,C5. The Salesman is looking for a path (tour) that passes through the 5 cities without passing through one city more than once. The Salesman wants to return to the city where he started his tour.

The distance between each city to another in table (2).
The solution can be written as 2→5→1→4→3→2. An itinerary that begins and ends at the same city and visits each city once is called a tour.

	C1	C2	C3	C4	C5
C1	0	132	217	164	158
C2	132	0	290	201	79
C3	217	290	0	113	303
C4	164	201	113	0	196
C5	158	79	303	196	0

How would you apply Tabu Search to solve this problem?

- Solution Representation: A feasible solution is represented as a sequence of nodes, each node appearing only once and in the order it is visited. The first and 1the last visited nodes are fixed to 1. The starting node is not specified in the solution representation and is always understood to be node 1.



- Initial Solution: A good feasible, yet not-optimal, solution to the TSP can be found quickly using a greedy approach. Starting with the first node in the tour, find the nearest node. Each time find the nearest unvisited node from the current node until all the nodes are visited.
- A neighborhood to a given solution is defined as any other solution that is obtained by a pair wise exchange of any two nodes in the solution. This always guarantees that any neighborhood to a feasible solution is always a feasible solution (i.e, does not form any sub-tour). If we fix node 1 as the start and the end node, for a problem of N nodes, there are $^{C-1}N_2$ such neighborhoods to a given solution. At each iteration, the neighborhood with the best objective value (minimum distance) is selected.

- Tabu List: To prevent the process from cycling in a small set of solutions, some attribute of recently visited solutions is stored in a Tabu List, which prevents their occurrence for a limited period. For our problem, the attribute used is a pair of nodes that have been exchanged recently. A Tabu structure stores the number of iterations for which a given pair of nodes is prohibited from exchange.
- Aspiration criterion: Tabus may sometimes be too powerful: they may prohibit attractive moves, even when there is no danger of cycling, or they may lead to an overall stagnation of the searching process. It may, therefore, become necessary to revoke tabus at times. The criterion used for this to happen in the present problem of TSP is to allow a move, even if it is tabu, if it results in a solution with an objective value better than that of the current best-known solution.
- Diversification: Quite often, the process may get trapped in a space of local optimum. To allow the process to search other parts of the solution space (to look for the global optimum), it is required to diversify the search process, driving it into new regions.
- Termination criteria: The algorithm terminates if a pre-specified number of iterations is reached.

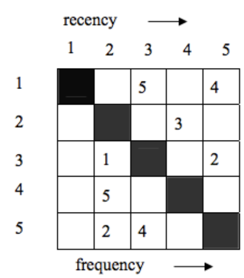


Figure 3: Tabu Structure [1]

i. What is the set of actions your Tabu search will use to build feasible solutions?
   City-Pair swapping.

ii. Assuming the initial solution 2->5->1->4->3->2 show the first 5 iterations of Tabu search. Assume a Tabu tenure t=2.

Suppose the initial solution to our problem is the shown in figure (3):

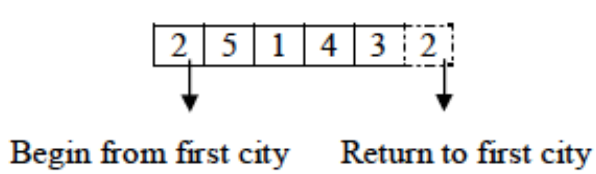


Figure (3)

Below table (3): Shows 7 iterations of the basic tabu procedure that contain the current solution, best solution, tabu structure and candidate list of moves.

Iteration	The current sol.	Best solution	Best solution (Represent the path between cities)	Tabu structure (Tabu tener)	Candidate list of moves			
					Swap between 2 cities		Objec-tive functi-on	Tabu status & best sol.
0	804	804	2→5→1→4→3→2	[0 0 0 0 0]	5	1	889	*
					3	5	946	
					5	1	889	
1	889	804	2→1→5→4→3→2	[0 0 0 0 3]	1	5	804	Tabu
					1	5	804	
					3	4	907	
2	804	804	2→5→1→4→3→2	[0 0 0 0 2]	5	4	1098	
					4	1	928	Tabu
					1	5	889	
3	928	804	2→5→4→1→3→2	[0 0 0 3 1]	5	1	1085	
					5	4	1062	
					3	5	786	Tabu
4	786	786	2→3→4→1→5→2	[0 0 0 2 0]	4	1	946	
					4	1	946	
					3	4	768	
5	768	768	2→4→3→1→5→2	[0 0 0 1 0]	4	5	964	
					1	4	737	Tabu
					5	4	964	
6	737	737	2→1→3→4→5→2	[0 0 0 0 0]	1	3	946	
					3	4	791	*
					3	1	946	
7	791	737	2→1→4→3→5→2	[0 0 0 3 0]				

Table (3)

Question 3. [25 Points] Simulated Annealing

a. Explain how SA achieves:
   Good meta-heuristics have a tradeoff between exploration/diversification and exploitation/intensification

   i. Search Diversification: sometimes accept local worsening moves, and sometimes, by restarts or reheats
   ii. Search Intensification: always accept improving moves

In Simulated annealing the temperature is reduced slowly, starting from a random search at high temperature eventually becoming pure greedy descent as it approaches zero temperature. The randomness should tend to jump out of local minima and find regions that have a low heuristic value; greedy descent will lead to local minima. At high temperatures, worsening steps are more likely than at lower temperatures.

b. What is the probability of accepting the following moves? Assume the problem is trying to maximize the objective function.

Current Evaluation	Neighbourhood Evaluation	Current Temperature
16	15	20
25	13	25
76	75	276
1256	1378	100

Current Evaluation (a)	Neighbourhood Evaluation (b)	Current Temperature (c)	Probability
16	15	20	0.95122942
25	13	25	0.61878339
76	75	276	0.99638337
1256	1378	100	3.38718773[1]

[1] This is the value returned from the SA accept function. The probability of accepting the move is 1. Either answer is correct.

b. The temperature parameter in Simulated Annealing determines the probability of accepting worse neighboring candidate solutions. Explain the impact high and, respectively, low temperatures have on the search behavior. SEE PART A

d. In Simulated Annealing. What effect have cooling schedules that drop the temperature very quickly (slowly)? Quickly- May result in suboptimal solutions. Slowly- allowing the method to search thoroughly at each temperature, and hence maximizing the chance of finding the global solution.

Question 4. [25 Points] Genetic Algorithm

a. What are the three basic components of any genetic algorithm?

b. Explain how GA achieves

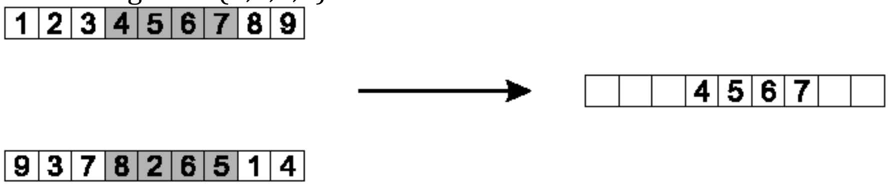
i. Search Diversification

ii. Search Intensification

c. Explain the difference between a genotypic representation and a phenotypic representation. Give an example of each.

d. Using an example, show why it is important to have a mutation operator in a genetic algorithm.

e. Show how PMX (Partially Mapped Crossover) can be used to perform crossover between the following two permutation parents using the highlighted parent segment (4,5,6,7).



f. Show how the n-point crossover operation is performed on the following two parents